

National Real-Time Manual Rotating Load Shedding Management Platform

Functional & Technical Specifications

Real-time coordination • Execution • Rotation • Monitoring • Transparency

1. Operational Context & Problem Statement

- Objective: maintain generation–load balance during generation deficit
- Trigger: Demand > Available Generation + Interconnections + operational margin
- Manual rotating load shedding is an emergency operational measure
- Current challenges: limited real-time visibility, delayed feedback, difficult rotation management, weak traceability
- Target: coordinated, measurable and transparent national process

2. Scope of the Platform

- J-1 forecast and real-time deficit assessment
- Creation and management of load-shedding orders
- Dispatching → CRC North/South → BCC allocation
- Real-time feeder monitoring and execution tracking
- Dynamic rotation and duration management
- Actual MW, elapsed time and energy-not-supplied calculation
- National monitoring, event logging and reporting

3. End-to-End Operational Process

FORECAST → OFFER/DEMAND BALANCE → DEFICIT IDENTIFICATION



LOAD-SHEDDING TARGET → ALLOCATION → EXECUTION



REAL-TIME MEASUREMENT → FEEDBACK → ADJUSTMENT



ROTATION → RESTORATION → REPORTING

Principle: every order must generate measurable, time-stamped feedback.

4. National Dispatching – Decision Layer

- Forecast demand and available generation
- Assess interconnection exchanges and reserve
- Identify expected or real-time deficit
- Determine required shedding power

$$P_{\text{deficit}} = P_{\text{Demand}} - P_{\text{Generation}} - P_{\text{Interconnections}}$$

Two modes:

- J-1 planned activation
- Real-time emergency activation

5. Dispatching → CRC North / CRC South

- National target is converted into regional targets
- Initial allocation can follow an agreed operational key
- Example: CRC North $\approx 2/3$; CRC South $\approx 1/3$
- Allocation must remain dynamically adjustable
- Every allocation/change is time-stamped and traceable

6. CRC → BCC Allocation

- CRC distributes its target among controlled BCCs
- Allocation based on real-time/reference load and network constraints
- Automatic consistency check:
$$\sum P_BCC = P_CRC$$
- BCC receives a clear target, deadline and operational context
- Platform shows target vs actual for every BCC

7. BCC – MT Feeder Selection

- Real-time list of source substations and MT feeders
- Feeder status: closed/open/unavailable
- Instantaneous load and estimated shed MW
- Priority and critical-load attributes
- Previous shedding duration and rotation history
- Selection objective: minimize $|\text{Target MW} - \text{Actual MW}|$ while respecting constraints

8. Dynamic Priority & Rotation Management

For every feeder, maintain:

- Current shedding duration
- Cumulative shedding duration
- Number of rotations
- Last shedding event
- Instantaneous power
- Operational priority

Rotation engine identifies feeders approaching the maximum duration and proposes replacement candidates.

9. Real-Time Execution & Additional Shedding

- Breaker OPEN event starts the feeder timer automatically
- Additional shedding orders can be issued at any time
- Platform identifies remaining deficit / target gap
- New feeders are proposed according to priority and availability
- Actual result is immediately returned to CRC and Dispatching

10. Real-Time Monitoring Dashboard – Dispatching

NATIONAL LOAD SHEDDING STATUS

Required: 300 MW Actual: 287 MW Gap: 13 MW

Feeders open: 27 Active BCCs: 7

CRC North: 191 MW CRC South: 96 MW

Maximum duration: 28 min

Energy Not Supplied: 1,245 MWh

Charts: Target vs Actual • MW by CRC/BCC • Open feeders • Duration • ENS

11. BCC Monitoring Screen

For each feeder:

- Feeder ID / Substation / BCC
- State: OPEN / CLOSED / UNAVAILABLE
- MW before and during shedding
- Opening timestamp
- Elapsed duration
- Remaining time to threshold
- Cumulative duration
- Rotation priority
- Alarm status

Example: F-125 | 13.4 MW | OPEN | 21:35 | limit 30 min

12. Closed-Loop Information Flow

FORWARD FLOW

Dispatching → CRC → BCC → MT Feeder

Order • Target • Allocation • Execution

RETURN FLOW

MT Feeder → BCC → CRC → Dispatching

Breaker status • MW • Duration • Energy • Events

The platform closes the operational loop and provides continuous visibility of execution.

13. Energy Not Supplied & Performance

For each feeder:

$$E = \int \text{PLS}(t) dt$$

$$\text{National ENS} = \sum \text{ENS}(\text{feeder})$$

Key KPIs:

- Target MW / Actual MW / Deviation MW
- Execution performance (%)
- Response time
- Number of feeders
- Average / maximum duration
- Number of rotations
- Energy Not Supplied (MWh)

14. Traceability & Transparency

Every event is recorded:

- Who initiated the order
- Time of order and execution
- Target and actual MW
- Feeders opened/closed
- Rotation events
- Alarm and threshold events
- Additional or reduced shedding orders
- Restoration actions

Result: complete reconstruction of every load-shedding event.

15. Target IT Architecture

DATA SOURCES

SCADA / RTU / EMS / BCC systems



INTEGRATION LAYER

OPC / ICCC / API / secure interfaces



LOAD SHEDDING MANAGEMENT PLATFORM

Database • Rules Engine • Allocation • Rotation • Alarms • Event Logger



APPLICATIONS

Dispatching Dashboard • CRC Dashboard • BCC Dashboard



REPORTING & ANALYTICS

Real-time KPIs • Historical trends • Event reports

16. Functional Requirements

F01 Create / validate / modify / cancel shedding orders

F02 Allocate targets Dispatching → CRC → BCC

F03 Monitor feeder states and MW in real time

F04 Automatic timers and threshold alarms

F05 Dynamic feeder priority and rotation support

F06 Additional shedding / reduction management

F07 Actual vs target performance calculation

F08 Energy Not Supplied calculation

F09 Full event logging and audit trail

F10 Dashboards and automated reports

17. Non-Functional Requirements

- 24/7 availability
- Near-real-time response
- High availability and redundancy
- Cybersecurity and role-based access
- Secure communications
- Complete audit trail
- Data integrity and time synchronization
- Resilience to communication loss
- Backup and disaster recovery
- Operational safety: controlled authorization of actions

18. Roles & Responsibilities

DISPATCHING

Decision • National target • Coordination • National monitoring

CRC NORTH / SOUTH

Regional allocation • Coordination • BCC supervision

BCC

Local execution • Feeder monitoring • Status feedback • Rotation

ADMINISTRATOR

Configuration • Users • Parameters • Audit • Maintenance

19. Implementation Roadmap

Phase 1 — Business & operational requirements

Phase 2 — Data inventory and interface specification

Phase 3 — Platform architecture and database

Phase 4 — Dispatching / CRC / BCC dashboards

Phase 5 — Real-time integration and alarm engine

Phase 6 — Rotation and optimization functions

Phase 7 — FAT / SAT / cybersecurity testing

Phase 8 — Pilot deployment

Phase 9 — National deployment and training

20. Expected Benefits

- National real-time visibility
- Faster execution of load-shedding orders
- Better accuracy between target and actual MW
- Controlled and equitable rotation
- Automatic duration supervision
- Complete traceability
- Transparent feedback from BCC to Dispatching
- Automatic calculation of ENS and KPIs
- Reliable post-event reporting

FROM MANUAL & FRAGMENTED CONTROL → TO A COORDINATED, MEASURED AND TRACEABLE NATIONAL SYSTEM

21. Target Operating Model

NATIONAL DISPATCHING



CRC NORTH / CRC SOUTH



BCCs



HT/MT SUBSTATIONS



MT FEEDERS



REAL-TIME MEASUREMENTS



BCC → CRC → DISPATCHING



MONITORING • FEEDBACK • ADJUSTMENT • ROTATION